```
Uganda, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.278             -33713.8            -33702.9                 0
     2 20263017             5000           0.279             -33697.1            -33683.7                 0
     3 20263018             5000           0.275             -33719.1            -33700.5                 0
     4 20263019             5000           0.276             -33725.1            -33717.6                 0
                                    file
 mcmc_uganda_step2_chain1_2026-10-19.Rdata
 mcmc_uganda_step2_chain2_2026-10-20.Rdata
 mcmc_uganda_step2_chain3_2026-10-20.Rdata
 mcmc_uganda_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                  male.hiv.mortality.multiplier.0       over.80.mortality.intercept.multiplier.male
                                            2.641                                             2.423
                  male.hiv.mortality.multiplier.1                         male.awareness.multiplier
                                            1.887                                             1.855
                                           trate.3            age.0.to.14.hiv.mortality.multiplier.1
                                            1.849                                             1.845
age.65.to.79.mortality.intercept.multiplier.female                                            trate.0
                                            1.731                                             1.720
           age.20.to.29.transmission.multiplier.3                                            trate.4
                                            1.712                                             1.695
  age.45.to.64.mortality.intercept.multiplier.male   age.65.to.79.mortality.intercept.multiplier.male
                                            1.659                                             1.639
           age.0.to.14.hiv.mortality.multiplier.2            age.20.to.29.transmission.multiplier.1
                                            1.632                                             1.628
                             log.OR.testing.slope            age.5.to.14.hiv.mortality.multiplier.0
                                            1.581                                             1.571
                          log.OR.testing.intercept                over.50.hiv.mortality.multiplier.2
                                            1.557                                             1.549
         age.25.to.49.hiv.mortality.multiplier.0         age.50.and.over.transmission.multiplier.3
                                            1.542                                             1.506
     age.45.to.64.mortality.slope.multiplier.male                                            trate.1
                                            1.501                                             1.435
                 male.hiv.mortality.multiplier.2            age.15.to.19.transmission.multiplier.2
                                            1.431                                             1.414
               over.50.hiv.mortality.multiplier.0
                                            1.412

Merged simset:
 location simulations total.weight                                          file
   Uganda        1000         1000 simset_uganda_step2_chains1234_2026-10-20.Rdata
```

Uganda step 2: likelihood

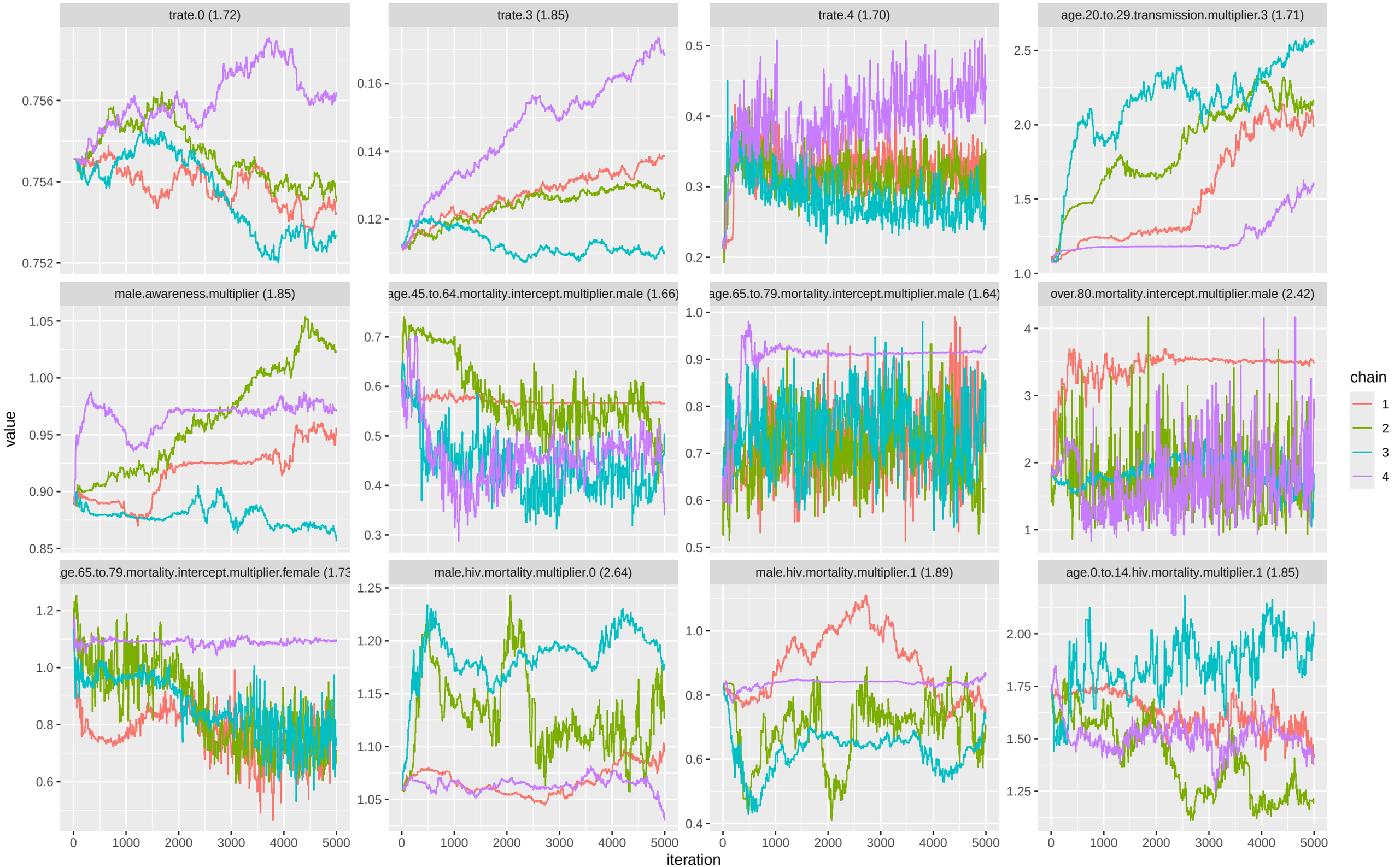
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

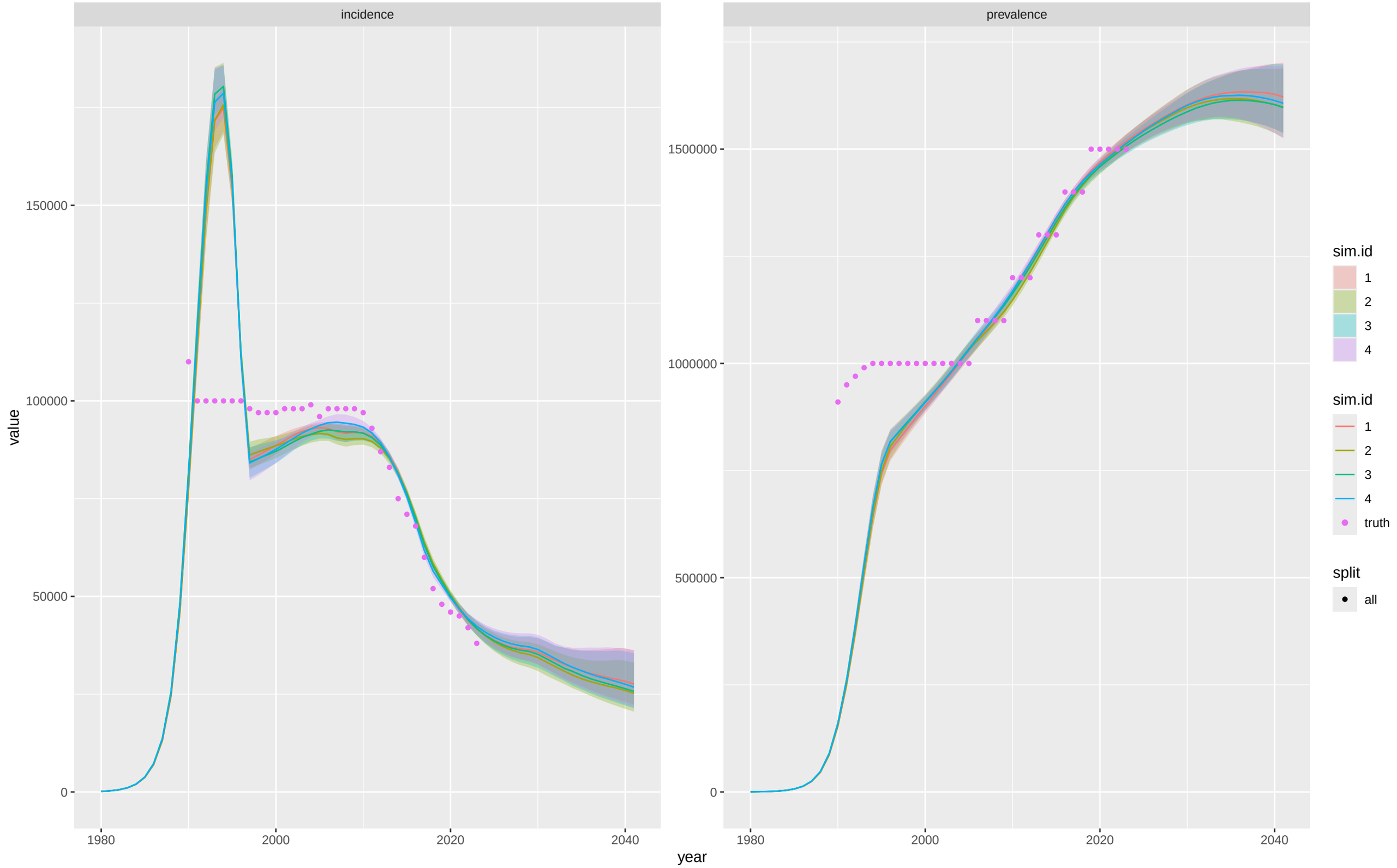
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

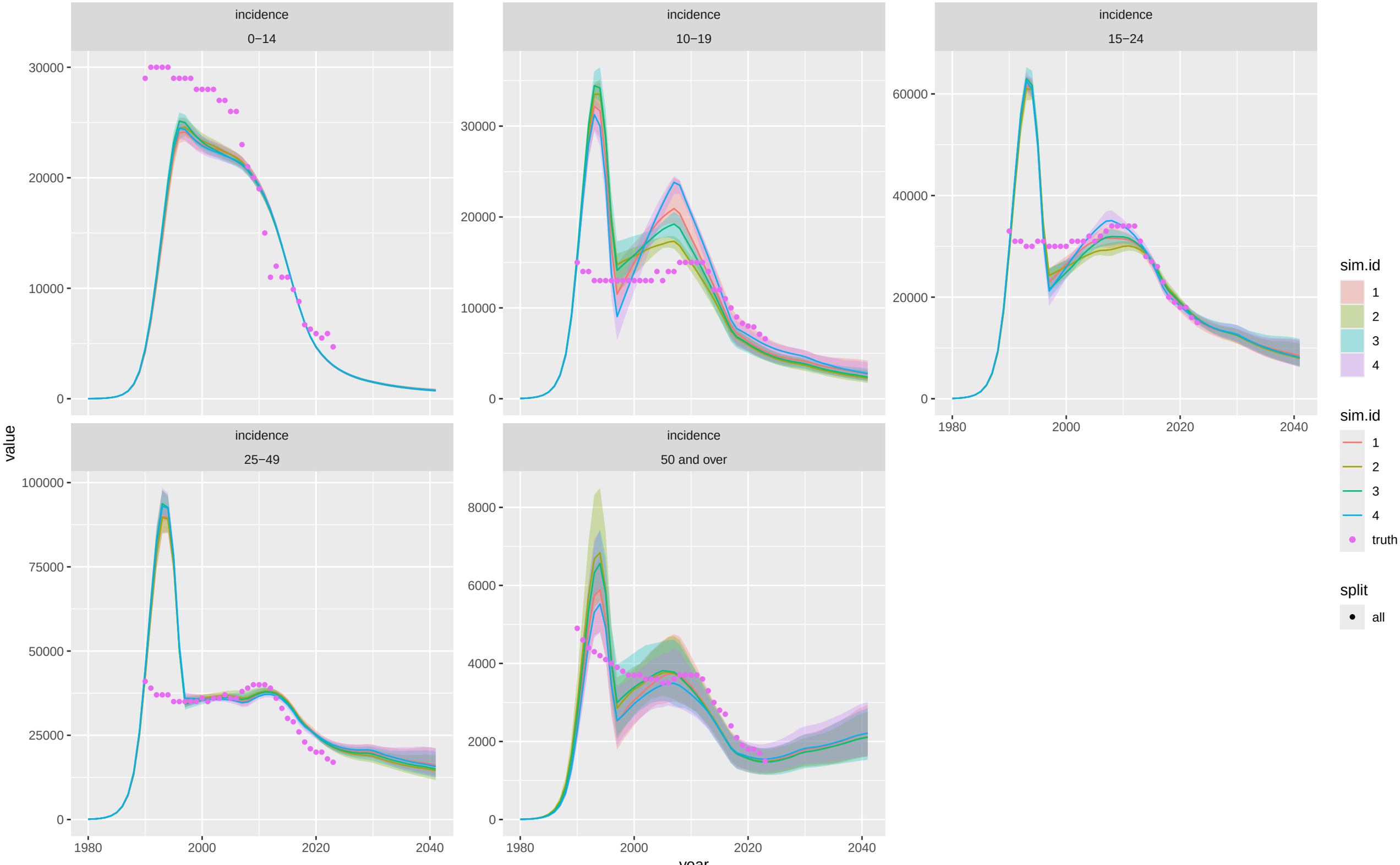
Uganda step 2: trace



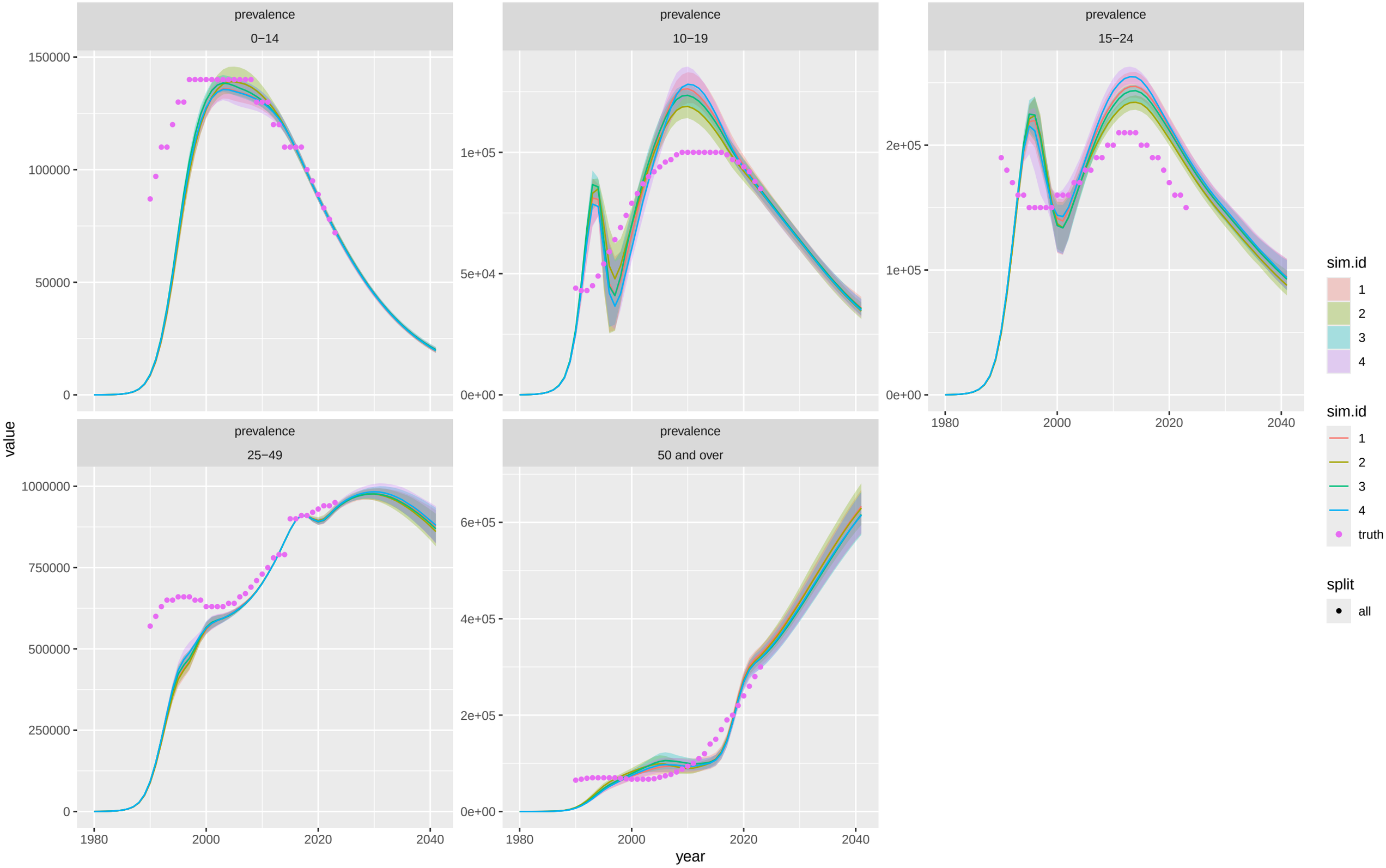
Uganda step 2: fit (one color per chain)



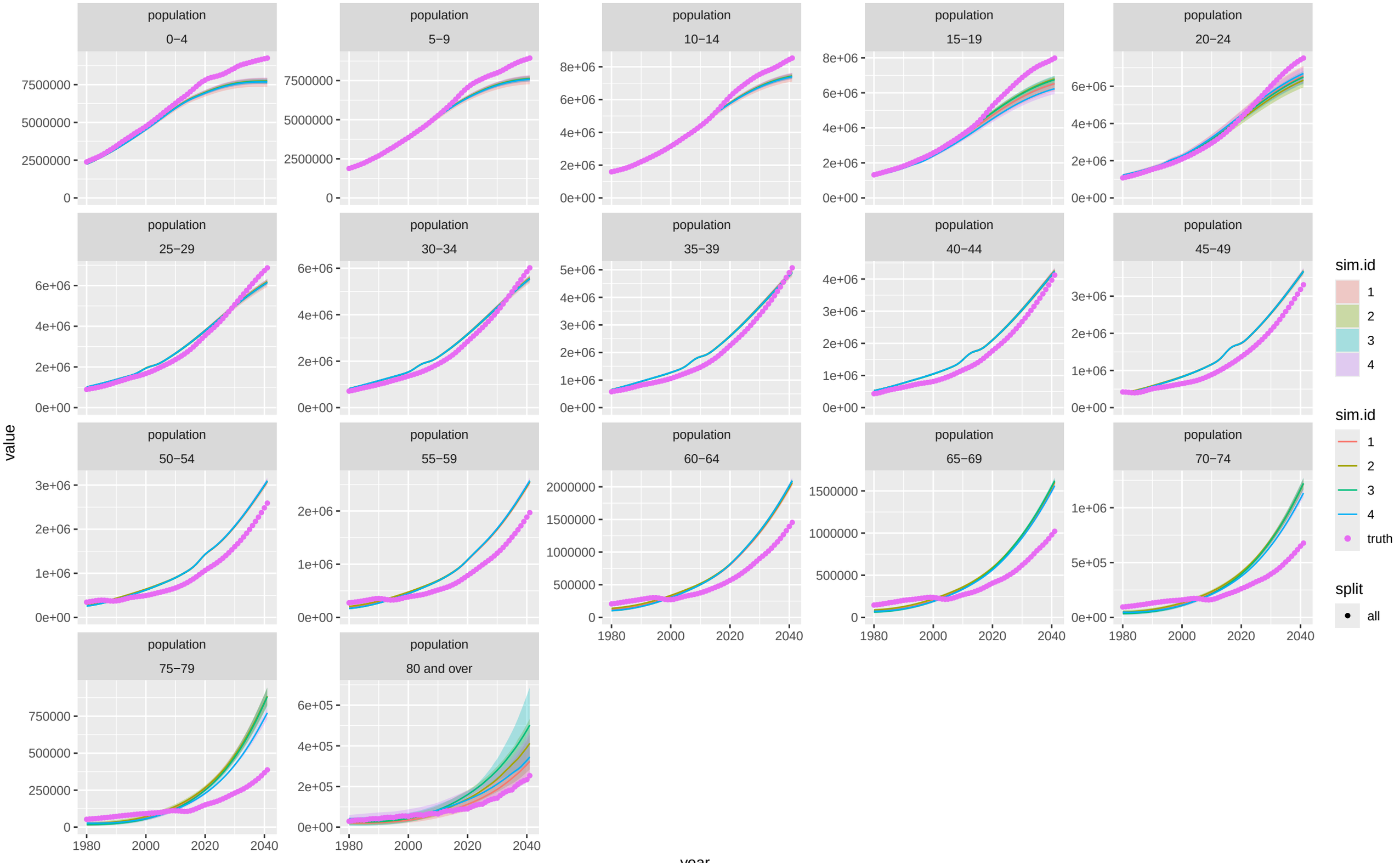
Uganda step 2: incidence.by.age (one color per chain)



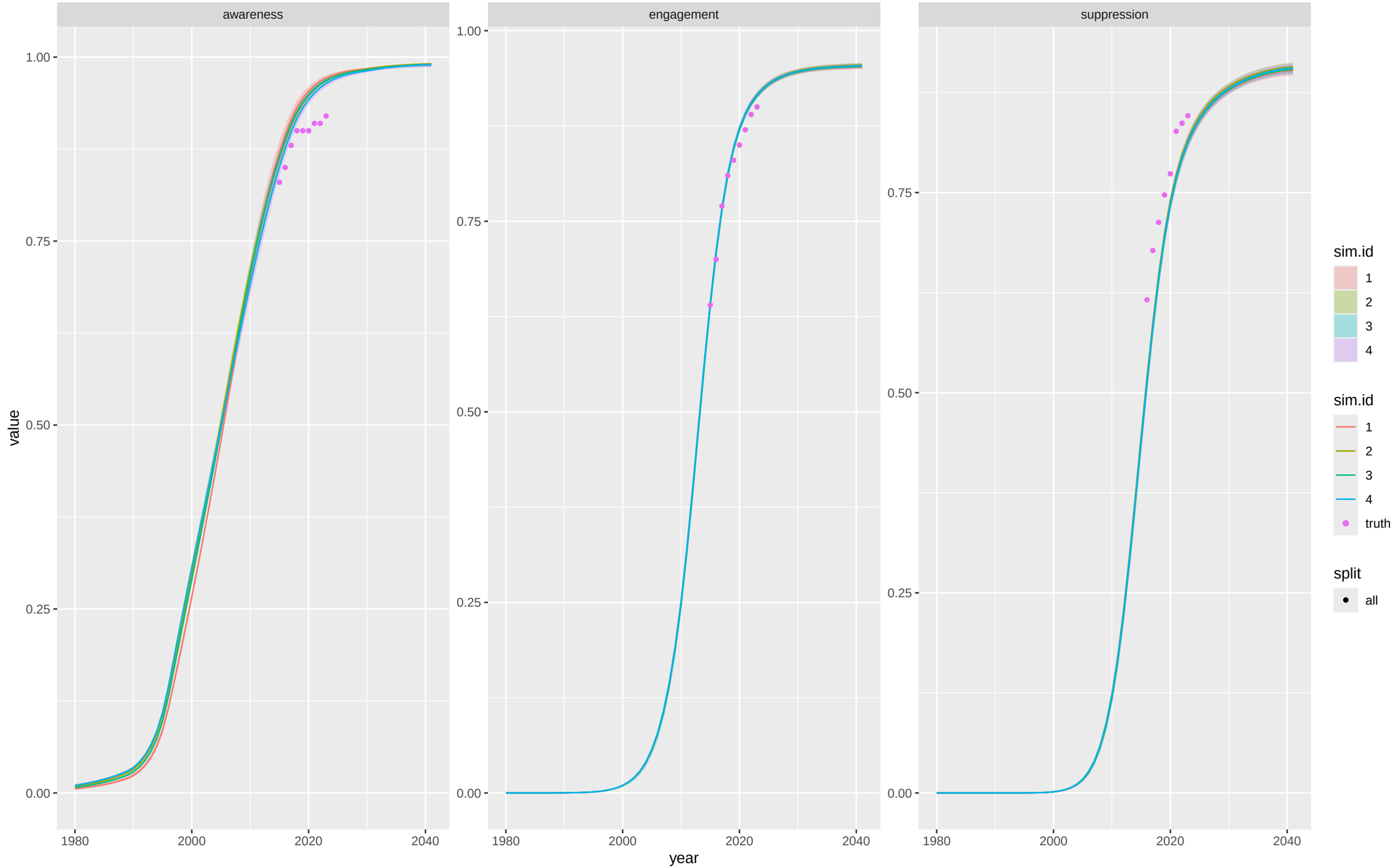
Uganda step 2: prevalence.by.age (one color per chain)



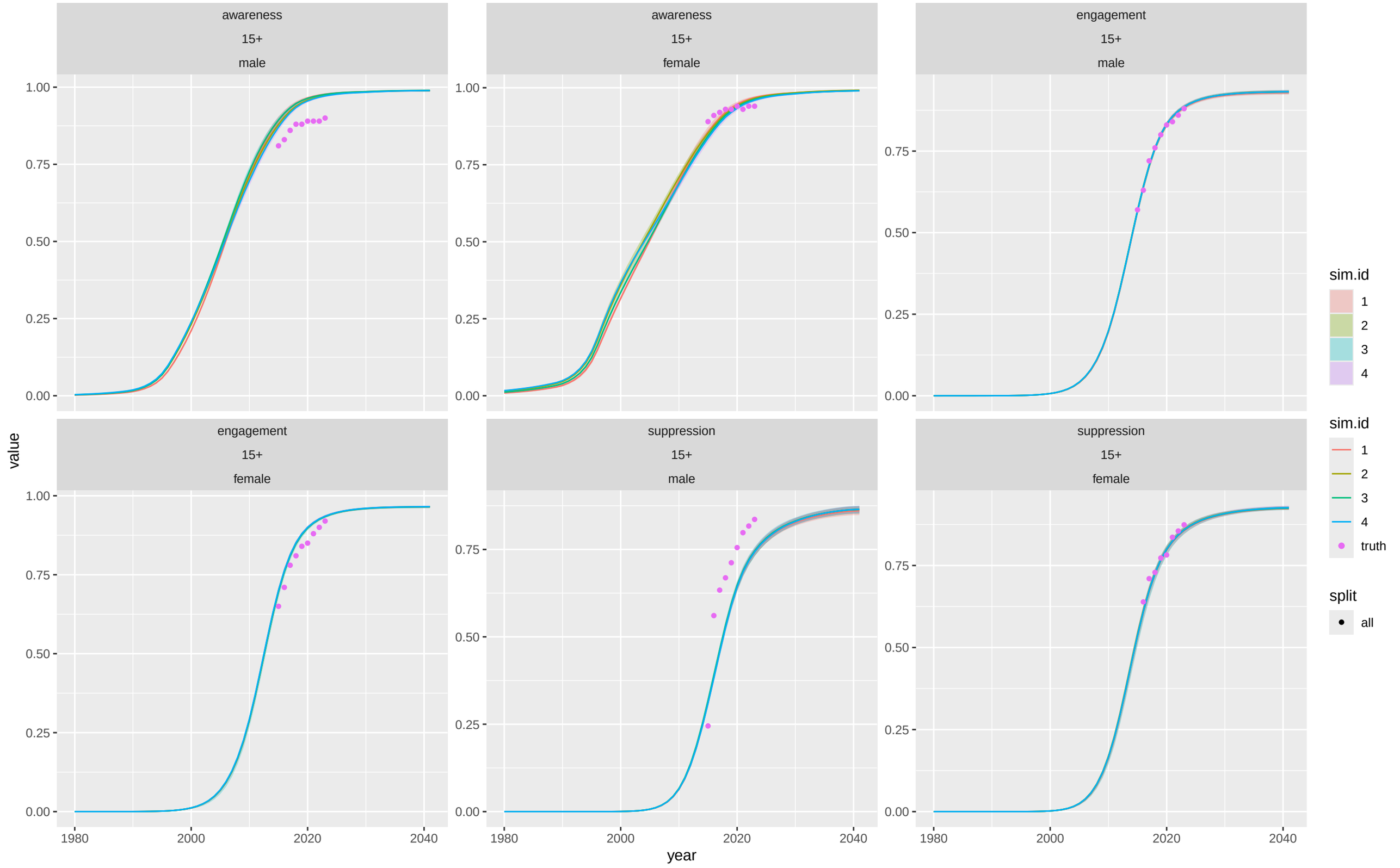
Uganda step 2: population.by.age (one color per chain)



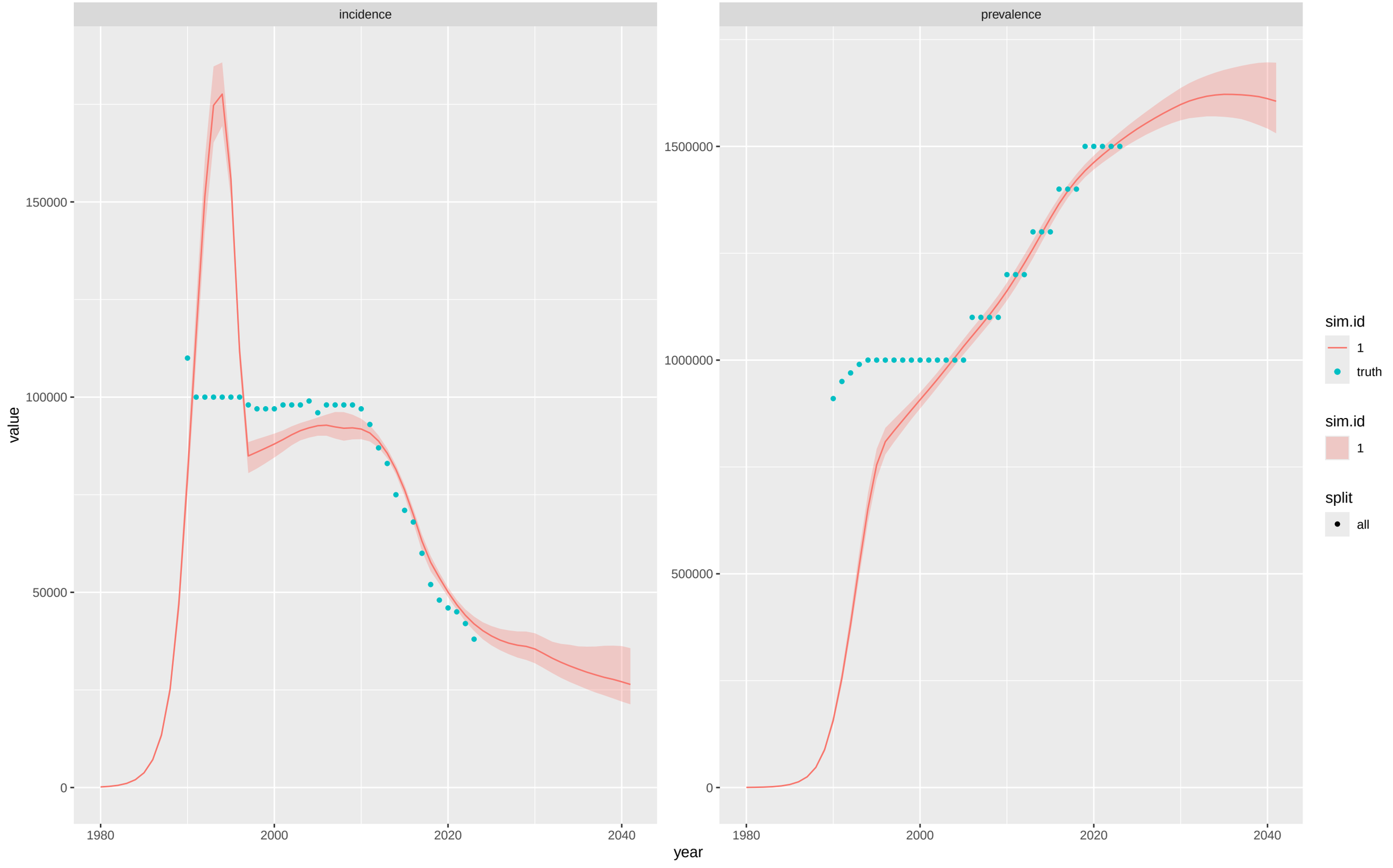
Uganda step 2: cascade (one color per chain)



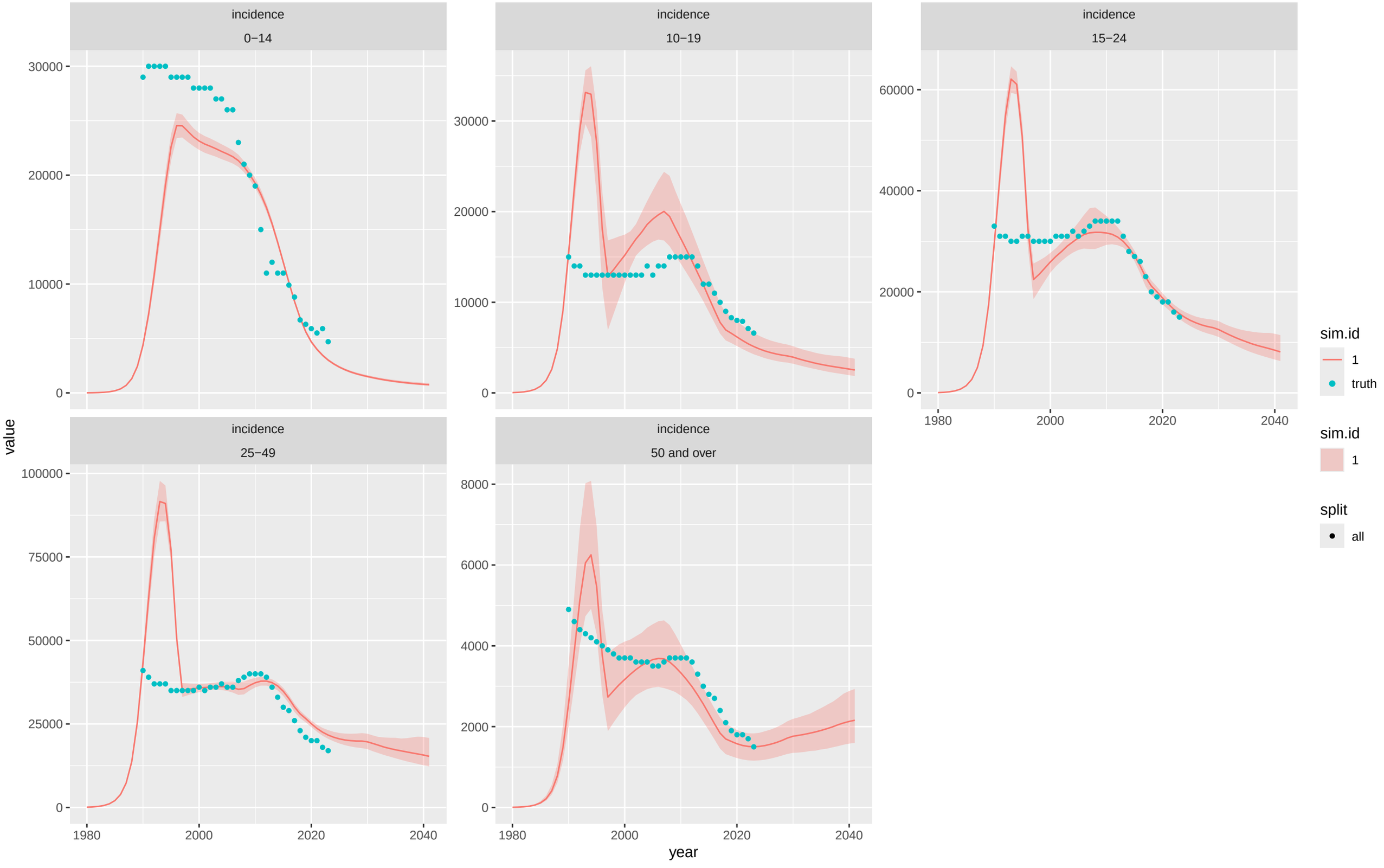
Uganda step 2: cascade.by.age.sex (one color per chain)



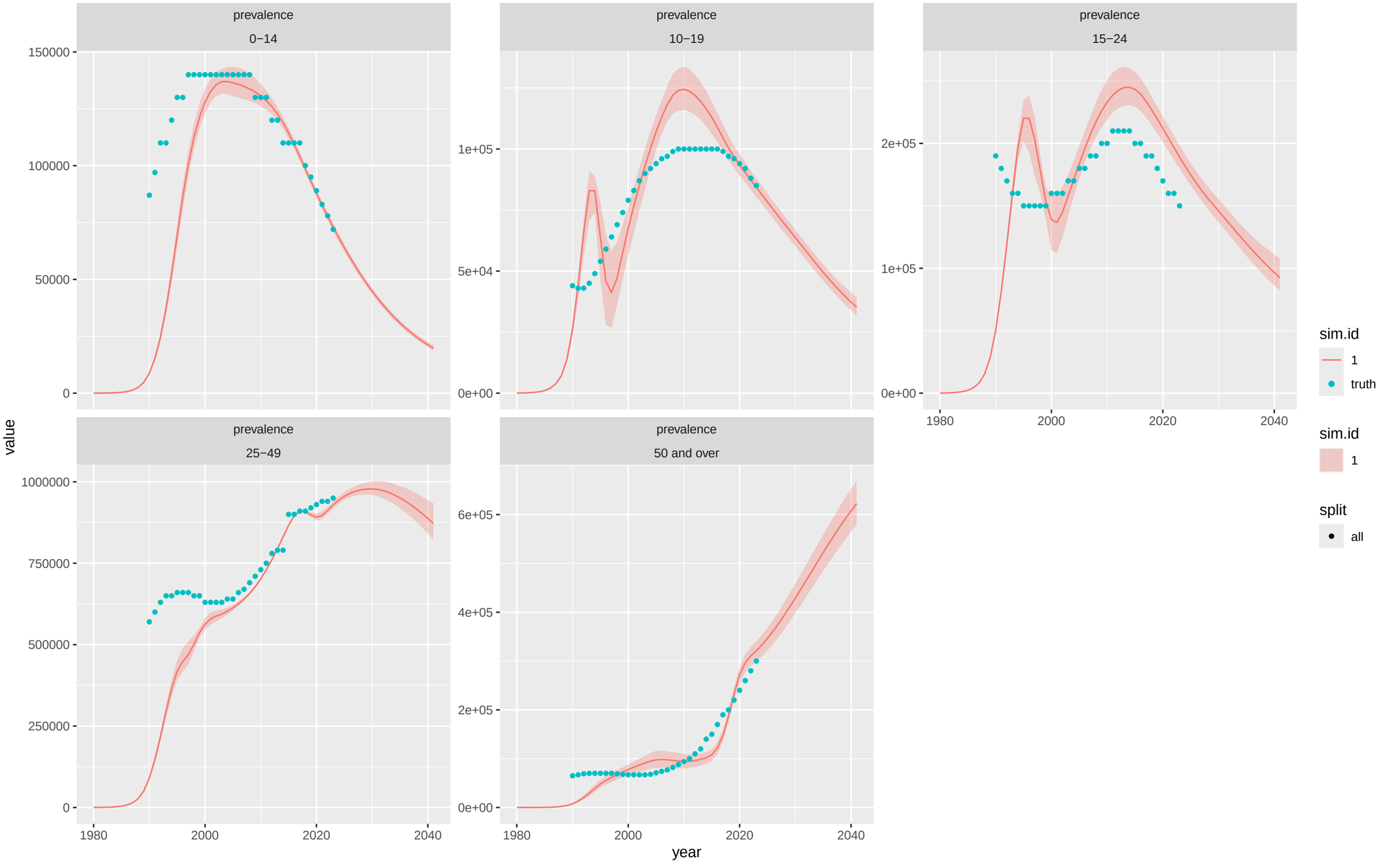
Uganda: fit (merged Step 2 simset)



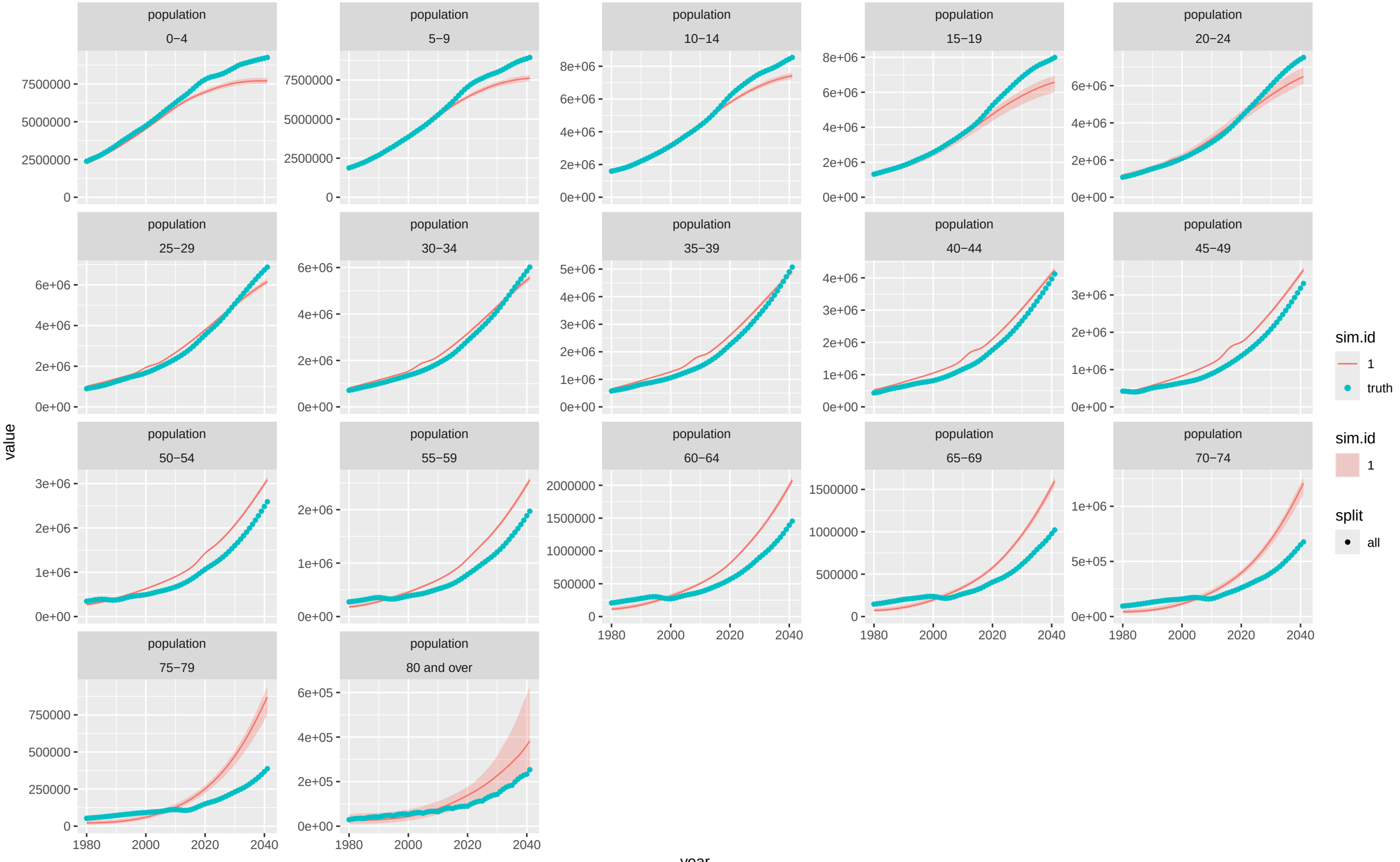
Uganda: incidence.by.age (merged Step 2 simset)



Uganda: prevalence.by.age (merged Step 2 simset)



Uganda: population.by.age (merged Step 2 simset)



Uganda: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

sim.id

1

truth

sim.id

1

split

all

Uganda: cascade.by.age.sex (merged Step 2 simset)

